

Autonomous Phenotyping

for Agentic EHR-Based Clinical Research Workflows

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Problem Statement

58%

of ICD-coded RA patients
confirmed on chart review

Raw EHR Data

ICD codes, notes, labs

Code-only Heuristics

Unreliable phenotyping

Inaccurate Cohorts

Missed / misclassified
cases

Our aim: automate the cohort building process.

Background & Related Work

Data Access

M4 / M3

NL→SQL on MIMIC-IV
(MIT Critical Data)

Ontologies

Vocabularies, OMOP, FHIR
interoperability layers

AI Scientists

ToolUniverse

Zitnik Lab — broad
clinical tool and skills library

AutoResearchClaw

Aiming Lab — end-to-end
research automation

EHR Phenotyping

MAP

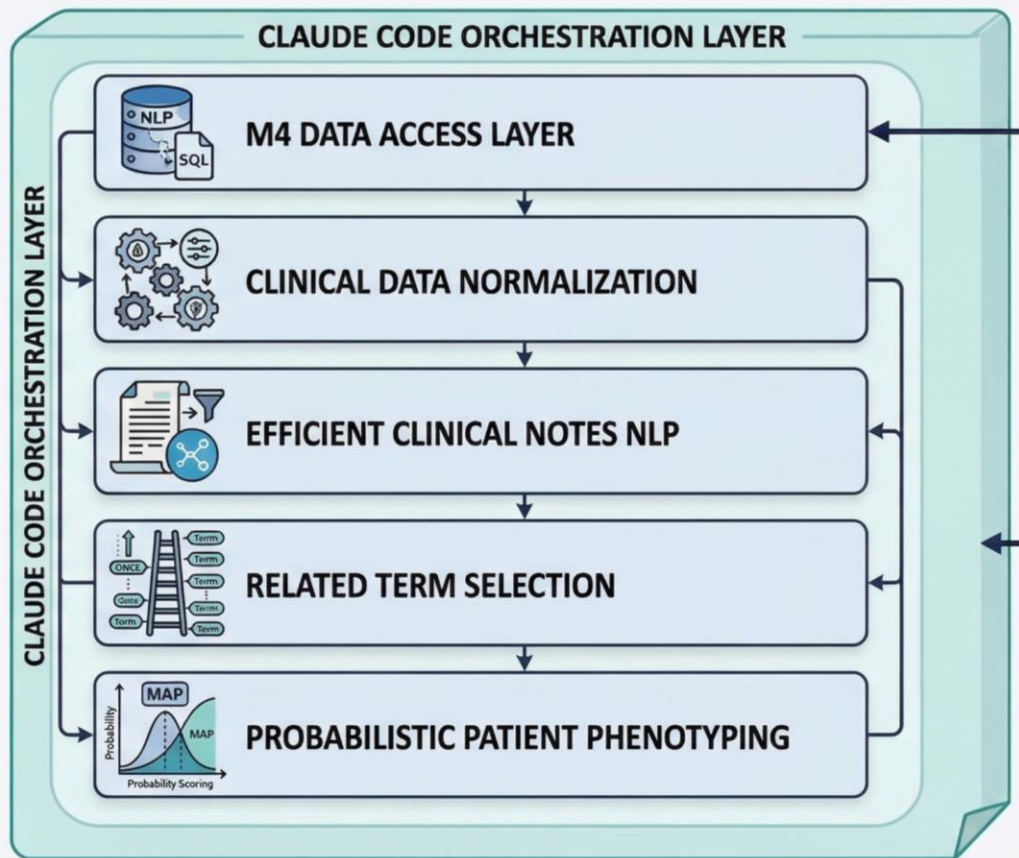
Probabilistic phenotyping
from codes + NLP

LATTE

Incident phenotyping
with temporal anchoring

The gap: no existing project automates the cohort building step end-to-end. We envision this project as a sub-agent or module pluggable into wider agentic clinical research pipelines.

Methodology



POC dataset

MIMIC-IV

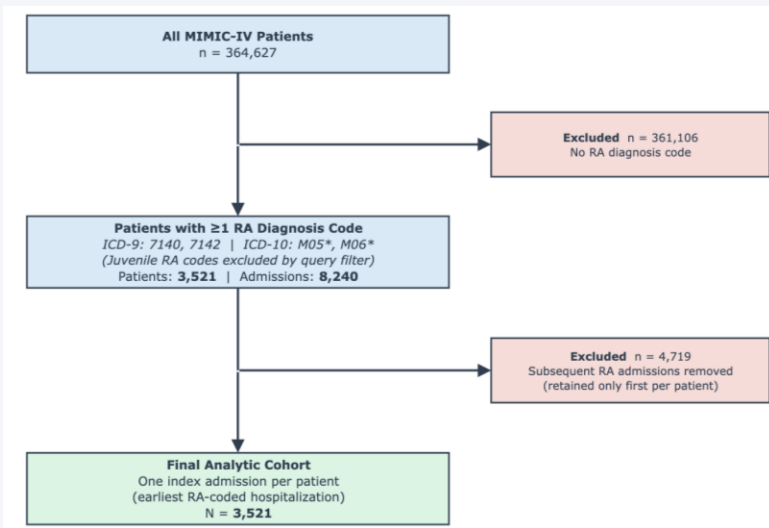
Widely accessible, comprehensive, contains both structured and unstructured data

Single Prompt → Full Pipeline

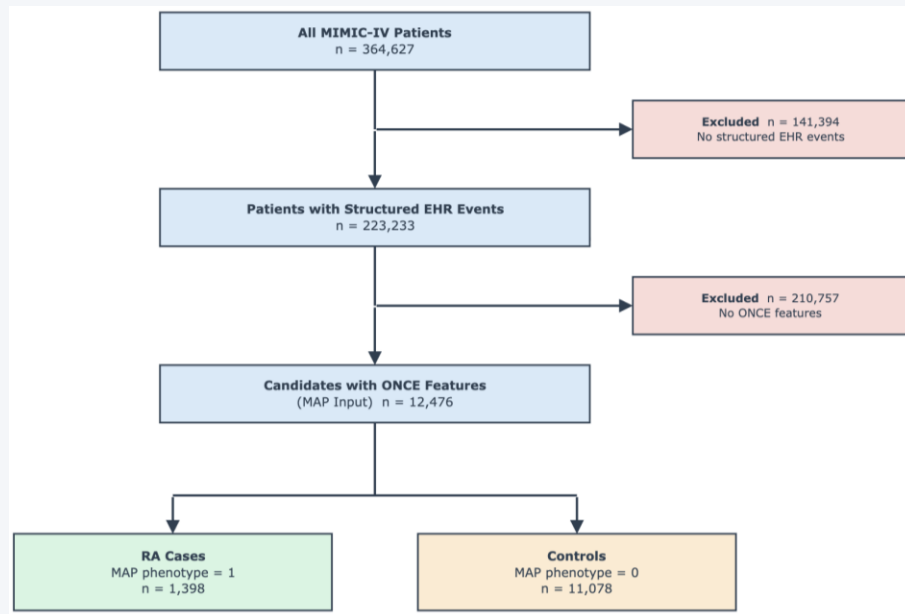
Research question → agent interview → preprocessing → NLP → phenotyping → characterization → self-evaluation

Results

Baseline Agent: MIMIC data access + basic cohort building
=> ICD-based filtering cohort:

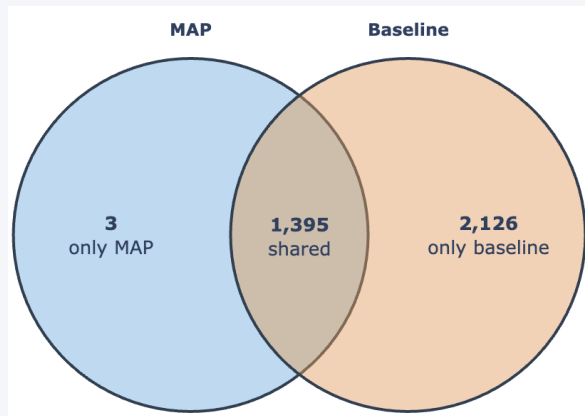


MAP Agent: baseline + preprocessing + NLP + phenotyping
=> multimodal phenotyping cohort:



1,398 << 3,521!

Results



Method	Final RA Cohort Size
MAP	1,398
Baseline	3,521
Shared	1,395
Only in MAP	3
Only in baseline	2,126

Key Points

- MAP improves specificity over baseline without introducing new cases, keeping only 40% of originally flagged patients
- Directionally consistent with MGB study, where 58% of multiple RA codes were true cases
- Confirms the agent is using MAP as intended: a multimodal high-precision filter that distinguishes true cases from rule-out diagnoses and billing artifacts

Next Steps and Challenges

Robust Evaluation of Agentic Phenotyping

Challenge

How to evaluate the agentic pipeline as a whole and the contribution of our skills without a ground truth?

Proposed Solution

Algorithm fidelity, ablation, paper reproducibility

LATTE for Temporal Phenotyping

Challenge

LATTE incorporates temporal information, which adds "messiness" such as temporal leakage

Proposed Solution

Coded guardrails to prevent temporal leakage

Improving Context Management

Challenge

Context rot with long sessions that may eventually reach model context limits

Proposed Solution

Native summarization or offloading tasks to sub-agents to reduce context

When an agentic system **chains validated algorithms**, what constitutes **sufficient evaluation**?

1

Algorithm Fidelity

Agent-invoked MAP = manual
MAP?

2

Ablation

Does each layer of the pipeline
add measurable gains?

3

Paper Reproducibility

Match pipeline results to
published cohorts with MAP

We are not proposing a new phenotyping method, but instead focusing on integrating and operationalizing existing phenotyping, preprocessing, and NLP into an autonomous workflow

Appendix

